

Eichler–Shimura defects of meromorphic period polynomials on the winding-class lattice

A computational session note

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Abstract

This note documents one session of work on the canonical-contour problem (item T1 of the pre-arXiv agenda): which winding classes of the two-segment contour give period polynomials of meromorphic modular forms satisfying the Eichler–Shimura relations $r|(1+S) = 0$ and $r|(1+U+U^2) = 0$. Everything below concerns the period polynomial of the *actual meromorphic form*, computed by direct numerical integration of f along the two-segment contour; no pole subtraction or projection enters any computation. The main outcomes: (i) two exact identities of the Hurwitz T -kernel — a reflection identity forcing $\mathbf{K}_T|(1+S) = 0$, and the fact that at integer s the kernel is a Bernoulli polynomial — which together reduce the whole problem to integer linear algebra with closed-form entries; (ii) the “the $(1+S)$ -defect of an i -pole is nonzero in every class” conjecture is *false*: an explicit class kills it (verified to 10^{-33}); (iii) for the weight-4 elliptic block, the classes satisfying *both* relations are completely characterized (Proposition 1): a nonempty rank-4 lattice coset, every weight in the defining equations derived by hand; (iv) in the clean class the original resonance proposition is restored and now hand-derived: the pole breaks exactly its resonant relation, by exactly its residue. Open items and all caveats are listed in §7.

1 Conventions

All conventions match `finite_contour_cocycles_short.tex` and its companion scripts. Fix even weight k and $n = k - 2$. For $f \in F_k$ (meromorphic modular of weight k for $\Gamma = \mathrm{PSL}_2(\mathbb{Z})$, meromorphic at the cusp), the L -integral is the two-segment contour integral

$$L^*(f, s) = e^{-\pi i s/2} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}(\tau, s) d\tau \right], \quad \tilde{k}(\tau, s) = \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(1-(k-s), \tau+1), \quad (1)$$

with $\gamma_{\tau_0}^S$ from $-1/\tau_0$ to τ_0 and $\gamma_{\tau_0}^T$ from $\tau_0 - 1$ to τ_0 . The period polynomial (overall $(2\pi i)^{n+1}$ dropped, as in the scripts) is

$$r_f(X, Y) = \sum_{\ell=0}^n c_\ell X^{n-\ell} Y^\ell, \quad c_\ell = (-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell+1), \quad (2)$$

equivalently $r_f = \int_{\gamma_S} f(X - \tau Y)^n d\tau + \int_{\gamma_T} f \mathbf{K}_T d\tau$ with the assembled kernel

$$\mathbf{K}_T(\tau; X, Y) = \sum_{\ell=0}^n \binom{n}{\ell} (-1)^\ell \tilde{k}(\tau, \ell+1) X^{n-\ell} Y^\ell. \quad (3)$$

Slash actions: $(P|S)(X, Y) = P(-Y, X)$, $(P|U)(X, Y) = P(X - Y, X)$, $(P|U^2)(X, Y) = P(-Y, X - Y)$. Define the two *defect operators*

$$D_S(P) := P|(1 + S), \quad D_U(P) := P|(1 + U + U^2).$$

For cusp forms and weakly holomorphic forms, $D_S(r_f) = D_U(r_f) = 0$ (the paper's §3; verified against Brown's quasi-periods and, via Haberland, against Petersson norms). The question is what replaces this for meromorphic f .

2 The winding lattice and the linear defect model

For meromorphic f the value of L^* depends on the homotopy class of the pair (γ^S, γ^T) in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, through winding data (X_S, X_T) : integers, one pair per pole, counting the net turns of each segment about each pole relative to a reference class (the paper's wall-crossing proposition). At the integer values $s = \ell + 1$ entering (2), one crossing of pole p by γ^S shifts r_f by $2\pi i R_p$, and one crossing by γ^T shifts it by $2\pi i Q_p$, where

$$R_p := \text{Res}_{\tau=p}(f(\tau)(X - \tau Y)^n), \quad Q_p := \text{Res}_{\tau=p}(f(\tau)\mathbf{K}_T(\tau; X, Y)). \quad (4)$$

Hence, writing $x = (x_p)$, $t = (t_p)$ for the winding integers of a class relative to a base class with polynomial $r_f^{(0)}$,

$$r_f(x, t) = r_f^{(0)} + 2\pi i \sum_p x_p R_p + 2\pi i \sum_p t_p Q_p, \quad (5)$$

and the defects D_S, D_U are *affine-linear in the winding integers*. Every question of the form “is there a class with $D_S(r_f) = 0$ and/or $D_U(r_f) = 0$ ” is therefore an integer linear-algebra question with coefficient vectors $D_S(R_p), D_U(R_p), D_S(Q_p), D_U(Q_p)$. This is the specialization of the paper's `eq:violation-residue` (verified there to 32 digits) used throughout.

3 Two exact identities of the T -kernel

Lemma 1 (Reflection). *For even k and integers $0 \leq m \leq n$,*

$$\tilde{k}(\tau, k - 1 - m) = -(-1)^m \tilde{k}(\tau, m + 1).$$

Proof. At $s = m + 1$ the phase is $e^{i\pi m} = (-1)^m$, so $\tilde{k}(\tau, m + 1) = \zeta(-m, \tau + 1) - (-1)^m \zeta(m + 2 - k, \tau + 1)$. At $s = k - 1 - m$ the two Hurwitz arguments are $1 - s = m + 2 - k$ and $1 - (k - s) = -m$, and the phase is $e^{i\pi(k-2-m)} = (-1)^m$ for even k ; so $\tilde{k}(\tau, k - 1 - m) = \zeta(m + 2 - k, \tau + 1) - (-1)^m \zeta(-m, \tau + 1)$, which is $-(-1)^m$ times the previous display. $\square$

Corollary 1. $\mathbf{K}_T|S = -\mathbf{K}_T$, hence $D_S(\mathbf{K}_T) = 0$ identically, and therefore:

(i) $D_S(Q_p) = 0$ for every pole p and every f : T -windings never move the $|(1 + S)$ -defect.

(ii) At $\tau_0 = i$ the S -segment degenerates to a point, so $r_f = \int_{\gamma^T} f \mathbf{K}_T$ and $D_S(r_f) = \int_{\gamma^T} f D_S(\mathbf{K}_T) = 0$: the S -relation holds automatically, for every $f \in F_k$ with no pole on the contour, in the $\tau_0 = i$ class.

Proof. In (3) substitute $(X, Y) \rightarrow (-Y, X)$ and re-index $\ell \rightarrow n - \ell$; the coefficient of $X^{n-\ell}Y^\ell$ in $\mathbf{K}_T|S$ is $\binom{n}{\ell}\tilde{k}(\tau, k-1-\ell)$, which by Lemma 1 equals $-(-1)^\ell\binom{n}{\ell}\tilde{k}(\tau, \ell+1)$, i.e. minus the coefficient in $\mathbf{K}_T$. Items (i), (ii) follow because D_S commutes with $\text{Res}_p(f \cdot)$ and with $\int_{\gamma^T} f \cdot$. $\square$

Corollary 1(ii) upgrades an empirical observation (all five test forms held $|(1+S)$ to $\sim 10^{-33}$ at $\tau_0 = i$, evidence item [E4] of the working ledger) to a theorem, and (i) decouples the problem: the S -defect is controlled by γ^S -windings alone, while γ^T -windings move only the U -defect.

Lemma 2 (Bernoulli form of the kernel). *At the integer points $s = m + 1$, $0 \leq m \leq n$, using $\zeta(-j, x) = -B_{j+1}(x)/(j+1)$,*

$$\tilde{k}(\tau, m+1) = -\frac{B_{m+1}(\tau+1)}{m+1} + (-1)^m \frac{B_{k-1-m}(\tau+1)}{k-1-m},$$

a polynomial in τ . In particular, for $k = 4$:

$$\tilde{k}(\tau, 1) = -B_1(\tau+1) + \frac{1}{3}B_3(\tau+1), \quad \tilde{k}(\tau, 2) = -B_2(\tau+1), \quad \tilde{k}(\tau, 3) = -\tilde{k}(\tau, 1),$$

$$\mathbf{K}_T = \tilde{k}(\tau, 1)(X^2 - Y^2) + 2B_2(\tau+1)XY. \quad (6)$$

Consequently *all* shift vectors (4) are residues of f against explicit polynomials: the canonical-class problem has closed-form integer data. (The visible $(X^2 - Y^2)$ and XY in (6) are each killed by $(1+S)$ — Corollary 1 made concrete.)

4 The weight-4 elliptic block: hand derivations and verdicts

Let $f = E_4\Delta/E_6^2 \in F_4$, the canonical block with a double pole at i (and no other poles mod Γ), Laurent data $a_{-2} = -1/(1728\pi^2)$ and the parity relation $a_{-1} = i a_{-2}$ (the paper's order-parity proposition at the S -fixed point).

4.1 Residue polynomials at the i -orbit

For a double pole, $\text{Res}_i(fg) = a_{-1}g(i) + a_{-2}g'(i)$ for holomorphic g . With $g = (X - \tau Y)^2$:

$$R_i = a_{-1}(X - iY)^2 - 2a_{-2}Y(X - iY) \stackrel{a_{-1}=ia_{-2}}{=} i a_{-2}(X^2 + Y^2). \quad (7)$$

By T -periodicity of f , $R_{i\pm 1}$ is (7) with $X \rightarrow X \mp Y$:

$$R_{i+1} = i a_{-2}((X - Y)^2 + Y^2), \quad R_{i-1} = i a_{-2}((X + Y)^2 + Y^2). \quad (8)$$

4.2 Defects of the shift vectors, by hand

Direct substitution gives

$$\begin{aligned} (X^2 + Y^2)|(1+S) &= 2(X^2 + Y^2), & ((X \mp Y)^2 + Y^2)|(1+S) &= 3(X^2 + Y^2), \\ (X^2 + Y^2)|(1+U+U^2) &= 4(X^2 - XY + Y^2), & ((X - Y)^2 + Y^2)|(1+U+U^2) &= 4(X^2 - XY + Y^2), \\ & & ((X + Y)^2 + Y^2)|(1+U+U^2) &= 8(X^2 - XY + Y^2). \end{aligned}$$

Writing $\sigma := 2\pi i R_i = -2\pi a_{-2}(X^2 + Y^2)$ for the unit on the S -side and normalizing the U -side by $D_U(2\pi i R_i)$, the shift table is:

generator	x_i	x_{i-1}	x_{i+1}	t_i	t_{i-1}	t_{i+1}
S -defect weight (units of σ)	2	3	3	0	0	0
U -defect weight (units of $D_U(2\pi i R_i)$)	1	2	1	7/12	7/12	19/12

The zeros in the T -columns of the S -row are Corollary 1(i). The integer entries 2, 3, 3 and 1, 2, 1 are the one-line computations above. The T -column entries are derived in §4.3; every entry of the table is therefore exact. (Each was independently confirmed by PSLQ at 34 digits, off-ray residuals $\lesssim 10^{-35}$.)

4.3 The T -columns, by hand

From (6), $\text{Res}_i(f\mathbf{K}_T) = a_{-1}\mathbf{K}_T(i) + a_{-2}\partial_\tau\mathbf{K}_T(i)$, with $\tilde{k}(\tau, 1) = \frac{1}{3}\tau^3 + \frac{1}{2}\tau^2 - \frac{5}{6}\tau - \frac{1}{2}$ and $B_2(\tau + 1) = \tau^2 + \tau + \frac{1}{6}$, so that $\tilde{k}(i, 1) = -1 - \frac{7}{6}i$, $\tilde{k}'(i, 1) = -\frac{11}{6} + i$, $B_2(i + 1) = -\frac{5}{6} + i$. Using $a_{-1} = ia_{-2}$,

$$Q_i = a_{-2} \left[(i\tilde{k}(i, 1) + \tilde{k}'(i, 1))(X^2 - Y^2) + 2(iB_2(i+1) + (2i+1))XY \right] = a_{-2} \left[-\frac{2}{3}(X^2 - Y^2) + \frac{7}{3}iXY \right].$$

Since $(X^2 - Y^2)|(1 + U + U^2) = 0$ while $XY|(1 + U + U^2) = X^2 - XY + Y^2$,

$$D_U(2\pi i Q_i) = -\frac{14}{3}\pi a_{-2}(X^2 - XY + Y^2) = \frac{7}{12}D_U(2\pi i R_i).$$

For the translates, the Bernoulli difference equations give $\mathbf{K}_T(\tau + 1) = \mathbf{K}_T(\tau) + (\tau^2 + 2\tau)(X^2 - Y^2) + 4(\tau + 1)XY$, whence (using 1-periodicity of f)

$$Q_{i+1} - Q_i = \text{Res}_i \left(f \cdot [(\tau^2 + 2\tau)(X^2 - Y^2) + 4(\tau + 1)XY] \right) = a_{-2} [i(X^2 - Y^2) + 4iXY],$$

$$Q_{i-1} - Q_i = -\text{Res}_i \left(f \cdot [(\tau^2 - 1)(X^2 - Y^2) + 4\tau XY] \right) = 0,$$

the second vanishing by the same parity cancellation ($a_{-1}(-2) + a_{-2}(2i) = 0$ and $a_{-1}(4i) + 4a_{-2} = 0$). Hence

$$\mu_U(t_{i-1}) = \frac{7}{12}, \quad \mu_U(t_{i+1}) = \frac{7}{12} + \frac{D_U(2\pi i(Q_{i+1} - Q_i))}{D_U(2\pi i R_i)} = \frac{7}{12} + 1 = \frac{19}{12},$$

completing the table. The equality $Q_{i-1} = Q_i$ — the repeated 7/12 — is thus a parity identity, not a numerical coincidence.

4.4 The clean class: the resonance proposition, restored

In the clean class ($\tau_0 = 0.3 + 1.4i$, straight segments, no pole enclosed), the measured defects of the actual r_f are

$$r_f|(1 + U + U^2) = 0 \text{ (residual } 5.6 \times 10^{-48}), \quad r_f|(1 + S) = -2\pi a_{-2}(X^2 + Y^2) = 2\pi i R_i, \quad (9)$$

the right-hand side now *derived* in (7) and equal to the measurement to 48 digits. Together with the mirror behaviour of ρ -pole forms (which in their clean classes hold $|(1 + S)$ and break $|(1 + U + U^2)$; see §5), this restores the original proposition in class-anchored form: *in the clean class, a pole on the fixed locus of an elliptic element breaks exactly its resonant relation, by exactly its residue polynomial*. What is genuinely class-dependent — and what the earlier “in every class” version missed — is quantified next.

4.5 The obstruction conjecture is false; joint classes exist

The earlier conjecture (ledger item [E5]) asserted the S -defect of an i -pole is nonzero in *every* class, based on sweeping only x_i (weight 2: even shifts of an odd base). The table refutes it: $\gcd(2, 3) = 1$, and indeed with base defect σ ,

$$x = (1, -1, 0) : \quad \sigma(1 + 2 - 3) = 0,$$

verified by direct high-precision recomputation of r_f in that class: $|(1 + S)|/\text{scale} = 7 \times 10^{-33}$ (the U -relation then breaks, as the table predicts). More is true: *both* relations can be killed simultaneously, and the solution set can be characterized completely.

Proposition 1 (Joint Eichler–Shimura exactness for the weight-4 block). *Let $f = E_4\Delta/E_6^2$, and take as reference the clean class, whose base defect pair is $(2\pi i R_i, 0)$ (9). Among the depth-one winding classes $(x_i, x_{i-1}, x_{i+1}; t_i, t_{i-1}, t_{i+1}) \in \mathbb{Z}^6$, those whose period polynomial satisfies both $r_f|(1 + S) = 0$ and $r_f|(1 + U + U^2) = 0$ are exactly the integer solutions of*

$$1 + 2x_i + 3x_{i-1} + 3x_{i+1} = 0, \quad 12x_i + 24x_{i-1} + 12x_{i+1} + 7t_i + 7t_{i-1} + 19t_{i+1} = 0. \quad (10)$$

This set is exactly the coset

$$(x; t) = (1, -1, 0; -1, 0, 1) + \mathbb{Z} \left\langle (0, 0, 0; 1, -1, 0), (0, 0, 0; -19, 0, 7), (-3, 2, 0; 1, 0, -1), (0, -1, 1; -1, 0, 1) \right\rangle, \quad (11)$$

a coset of a rank-4 sublattice of $\mathbb{Z}^6$. In particular Eichler–Shimura exactness holds in infinitely many classes, and exactness alone does not select a contour.

Proof. By the linear model (5) (the wall-crossing law `eq:violation-residue` of the paper, specialized to integer s), the two defects of the class $(x; t)$ are affine-linear in the winding integers with shift vectors D_S, D_U of $2\pi i R_p, 2\pi i Q_p$ at $p \in \{i, i \pm 1\}$. All twelve weights are derived above: the S -row is $(2, 3, 3; 0, 0, 0)$ in units of $\sigma = 2\pi i R_i$ (§4.2 for the integer entries via (7)–(8); the T -entries vanish by Corollary 1(i)), and the U -row is $(1, 2, 1; \frac{7}{12}, \frac{7}{12}, \frac{19}{12})$ in units of $D_U(2\pi i R_i)$ (§4.2 and §4.3). Both defect components lie on the fixed rays $\mathbb{C}(X^2 + Y^2), \mathbb{C}(X^2 - XY + Y^2)$, so vanishing of the defect pair is equivalent to the two scalar equations: base 1 plus the S -row for the first; base 0 plus the U -row, cleared of denominators ($\times 12$), for the second — i.e. (10). Diophantine solution. The integer solutions of the homogeneous equation $2a + 3b + 3c = 0$ are exactly $\alpha(3, -2, 0) + \gamma(0, -1, 1)$: reducing mod 3 forces $3 \mid 2a$, hence $3 \mid a$, and the parametrization follows. A particular solution of the first equation of (10) is $x^{(0)} = (1, -1, 0)$. Writing $N := x_i + 2x_{i-1} + x_{i+1}$, the second equation reads $7t_i + 7t_{i-1} + 19t_{i+1} = -12N$, with particular solution $t = N(1, 0, -1)$ (because $7 - 19 = -12$) and homogeneous solutions $s(1, -1, 0) + u(-19, 0, 7)$ (reducing mod 7 forces $7 \mid 19t_{i+1}$, hence $7 \mid t_{i+1}$). Since $N(x^{(0)}) = -1$, $N(-3, 2, 0) = 1$, and $N(0, -1, 1) = -1$, assembling particular-plus-kernel in x with the matching $t = N(1, 0, -1)$ gives exactly the coset (11); the four displayed kernel vectors generate the *full* kernel lattice, since any homogeneous solution, after subtracting multiples of the third and fourth, has x -part zero and is then a combination of the first two. Each generator checks against (10) directly, e.g. $(-3, 2, 0; 1, 0, -1)$: $-6 + 6 = 0$ and $-36 + 48 + 7 - 19 = 0$; the particular solution: $1 + 2 - 3 = 0$ and $12 - 24 - 7 + 19 = 0$. $\square$

(A float box search found 66 of these solutions with all windings in $[-3, 3]$, each consistent with (10) — the numerical shadow of the proposition. The inputs a reader may wish to interrogate: the wall-crossing law, proven in the paper and verified to 32 digits; the parity $a_{-1} = ia_{-2}$, the paper’s order-parity result at the S -fixed point; and the clean-class base pair, measured to 48 digits, (9).)

5 The ρ -blocks and a generic pole

The same machinery was run on E_6/j ($k = 6$, triple pole at $\rho = e^{2\pi i/3}$), $E_8/j = \Delta/E_4$ ($k = 8$, simple pole at ρ), and the generic-pole form $E_6/(j - j(1.1i))$ ($k = 6$). Base class $\tau_0 = i$ throughout (which holds $|(1 + S)|$ identically by Corollary 1(ii)); windings allowed at the poles adjacent to the contour (“depth one”: the poles and their T -translates and S -images next to the fundamental domain), boxes $|x| \leq 3$.

E_6/j . All U -defects lie on one real ray with PSLQ-exact weights

$$\mu_U : \quad x_\rho : -11, \quad x_{\rho+1} : -3, \quad t_\rho : -\frac{8}{5}, \quad t_{\rho+1} : -\frac{48}{5}, \quad \text{base} = 1.$$

Killing U alone is feasible: $3 \cdot (-3) + 1 \cdot (-\frac{8}{5}) + (-1) \cdot (-\frac{48}{5}) = -1$, i.e. $(x, t) = ((0, 3), (1, -1))$, verified to 4.1×10^{-31} (the S -relation then breaks, weight 6.0). Killing *both* failed in the depth-one window: the two S -defect vectors $D_S(R_\rho), D_S(R_{\rho+1})$ are $\mathbb{Z}$ -independent (no rational ray), pinning $x_\rho = x_{\rho+1} = 0$; the T -windings alone then need $\frac{8}{5}t_1 + \frac{48}{5}t_2 = 1$, i.e. $8t_1 + 48t_2 = 5$ — impossible mod 8. *Caveat: this is an obstruction of the truncated window, not (yet) of the full winding lattice; deeper windings refine the reachable lattice and may dissolve it.* No impossibility proposition is claimed here or anywhere in this note: the mod-8 argument is exact only within the depth-one window and rests on the PSLQ-identified weights (not yet hand-derived, unlike the $k = 4$ table). The honest general statement is:

Question. For which blocks does the winding lattice reach joint exactness — equivalently, for which pole orbits does the defect coset contain the origin? The weight-4 i -block does (Proposition 1); the depth-one window of E_6/j does not; the full lattice for E_6/j , Δ/E_4 , and the generic pole is undecided.

Δ/E_4 ($k = 8$). The U -defect space is genuinely multi-dimensional (no single ray; $\dim \ker(1 - U) = 3$ on degree-6 forms), rank 7 of 10 generators; no joint solution in the $|x| \leq 2$ window. Inconclusive.

$E_6/(j - j(1.1i))$ (**generic pole**). Base class: S -defect = 0 (Corollary 1(ii)), U -defect $\neq 0$. Each relation is separately killable (this reproduces the earlier routed-contour observations); rank 7 of 12 generators, real solutions of the joint system exist, but no *integer* joint solution appeared with the five free windings in $[-3, 3]$. Inconclusive; see §7.

6 The corrected picture

For meromorphic f the period polynomial is canonically a *coset*: the set (5) as (x, t) ranges over the winding lattice. The defect pair $(D_S(r_f), D_U(r_f))$ is an affine-linear image of that lattice; single defect values are class data, the coset is the invariant. The numerical record then organizes itself:

- classes on the axis $D_S = 0$ always exist — e.g. $\tau_0 = i$, now a theorem (Cor. 1(ii));
- classes on the axis $D_U = 0$ existed in every case tested (clean class for the i -block, (9); routed classes for the $1.1i$ -forms; the explicit solution for E_6/j);
- whether the coset contains the *origin* — both relations at once — is the only genuinely open obstruction. For the weight-4 block it does (Proposition 1, a rank-4 lattice of such classes); for the ρ -blocks and the generic pole it is undecided at depth one.

Two inequivalent notions of “canonical class” emerge, and they disagree whenever the residue is nonzero:

1. **The clean class:** geometrically distinguished, essentially unique, defines the L -integral for any meromorphic form; the Eichler–Shimura relations hold up to an explicit anomaly which *is* the residue polynomial (9) — canonical pole data in closed form.
2. **Eichler–Shimura–exact classes:** exact relations, but (where they exist) a full lattice of them, each with a different L^* ; exactness alone does not select a contour, and per-block existence is an arithmetic question.

Which notion the paper should adopt is a judgment call (the natural tie-breakers within (2) being central-value crossing-invariance and the Haberland/Petersson pairing); the data of this note supports stating (1) as the definition and (2) as structure theory of the coset.

7 Open items and caveats

1. **Hand derivation of the E_6/j weight table** $(-11, -3, -8/5, -48/5)$: pending. (The $k = 4$ table is now fully derived, §4.3.) This would make the mod-8 window obstruction of §5 unconditional within its window.
2. **Depth ≥ 2 windings** for $E_6/(j - j(1.1i))$ and E_6/j : untested (a script `t1_depth2.py` exists, never run). All “no joint solution” statements above are *window-limited*, not proofs.
3. **Speculative** (flagged as such; not verified): since $\mathrm{PSL}_2(\mathbb{Z}) \cong \mathbb{Z}/2 * \mathbb{Z}/3$, the obstruction to an Eichler–Shimura–exact class should be a pair of torsion classes in $H^2(\mathbb{Z}/2; \cdot) \oplus H^2(\mathbb{Z}/3; \cdot)$ with coefficients in the winding module; free orbits (generic poles) would then carry no obstruction, elliptic-orbit poles possibly a finite one. Two identified gaps: the precise identification of the two H^2 components with the two defects, and the module structure of the T -shift vectors Q_p . This predicts a depth- ≥ 2 joint solution for the generic-pole form — a falsifiable test, not yet run.
4. The “66 solutions” count is a float box search cross-checked against the exact ray equations; the exact statement is the Smith-normal-form solution lattice of §4.
5. The $k = 8$ (Δ/E_4) defect geometry (multi-dimensional U -space) is unmapped.

A Numerical methods and scripts

All computations: `mpmath` at 34 significant digits, conventions imported from `wip_meromorphic_periods.py`. Period polynomials of the actual forms by direct `mp.quad` of (1) at $s = 1, \dots, k-1$ (two-segment contour; $\tau_0 = i$ degenerate class or $\tau_0 = 0.3 + 1.4i$ clean class). Residue vectors (4) by circle sums (radius 0.02, 64 points; Hurwitz zeta evaluated directly for the Q_p). Rational identification by PSLQ; integer feasibility by exact Smith normal form on the cleared-denominator ray equations; joint searches by kernel-aware enumeration cross-checked at full precision. Session scripts (scratchpad, in order): `t1_lattice.py` (linear model, first sweeps), `t1_structure.py` (exact ray analysis, PSLQ weights, falsification of [E5], joint-kill box search), `t1_exact.py` (Smith-normal-form verdicts, extended ρ -reps, kernel-aware search), `t1_depth2.py` (depth-2 test, written, *not run*). Headline residuals: $\mathbf{K}_T|(1+S)$ machine-exact (identity); [E5]-falsifying class 7×10^{-33} ; clean-class U -relation 5.6×10^{-48} ; E_6/j U -kill 4.1×10^{-31} ; PSLQ off-ray residuals $\lesssim 10^{-35}$.